2.1 Stakeholder analysis and conflict resolution

Stakeholders	Primary Interests
Principal	Simple system for non-IT teachers; students must not see marks before HoD check.
Head of Department (HoD)	See every mark in the department before release; consistent late penalty (10% a day for 3 days, then zero); release approved marks.
IT Officer	Fully offline operation; strong password protection; good performance with 800 students.
Student	Ability to resubmit before deadline; written feedback; late work not automatically rejected.
Lecture	Quick release of marks after marking; ability to grant individual extensions.
Administrator	Manage users, departments, courses, enrolments and lecturer assignments.

Conflicts:

Conflict ID	Statements	Issue	Resolution
C1	Principal: Students must not see any mark until it has been checked by the HoD. Lecture: Students should get their marks as soon as I have marked the work.	When should students see marks?	The principal's requirement takes precedence because it represents the school's assessment-control policy. Therefore: A lecturer may mark a submission and provide feedback, but the mark remains hidden from the student until the HoD has moderated and released it.
C2	Principal: System must be simple. Student: resubmission, written feedback. Lecture: gives extensions. HoD: Late submission penalty.	Every requested feature adds complexity for non-expert users.	All essential stakeholder functionality will be retained. The system will address the usability concern through role-specific interfaces, clear workflows, input

			validation, and automation of rules such as late-penalty calculation.
C3	HoD: after three days zero. Student: late work should not simply be rejected.	Zero mark or rejection after three days?	Late submissions are accepted up to 3 days after the deadline. A penalty of 10% of the maximum mark is applied for each day late. After 3 days the mark is set to zero. Students can still submit (and resubmit) within the late window.
C4	IT: system must work offline. Brief: single computer or local network.	Is multi-computer access needed?	No internet is needed at any time. The database file is local to one installation and multi-machine access is not supported in v1.0.

Ambiguities:

Ambiguity ID	Statements	Issue	Resolution
A1	HoD: lose 10% of the available mark for each day late.	10% of what: the maximum mark, or the mark earned?	10% of the assignment's maximum mark per day, subtracted from the raw mark and not below zero. 'Available mark' means the marks on offer. A fixed deduction is also easier to explain to students than a percentage of their own mark.
A2	HoD: review and moderate marks for all courses in the department.	Does moderation mean read-only checking or the power to change marks?	The HoD can approve, adjust (with a mandatory reason) or return a mark to the lecturer.

			Moderation without power to change a mark has little purpose, and every change is audited.

2.2 Software Requirements Specification (SRS)

Functional Requirements

ID(FR)	Requirement	Priority	Verification
FR-01	The system shall authenticate users using a username and password.	Must	Test valid and invalid login attempts.
FR-02	The system shall restrict operations according to the authenticated user's role.	Must	Test permitted and prohibited actions for all roles.
FR-03	The administrator shall create, update and deactivate user accounts.	Must	Functional tests for account management.
FR-04	The administrator shall create and manage departments and courses.	Must	Test course and department operations.
FR-05	The administrator shall assign lecturers to courses and enroll students in courses.	Must	Test assignments and enrolments.
FR-06	A lecturer shall create assignments with a title, description, deadline and maximum mark.	Must	Create assignments with valid and invalid data.
FR-07	A lecturer shall grant an individual student an extension by specifying a revised deadline.	Must	Test extension creation and deadline updates.
FR-08	A student shall view the courses in which they are enrolled and the assignments	Must	Test visibility using different enrolments.

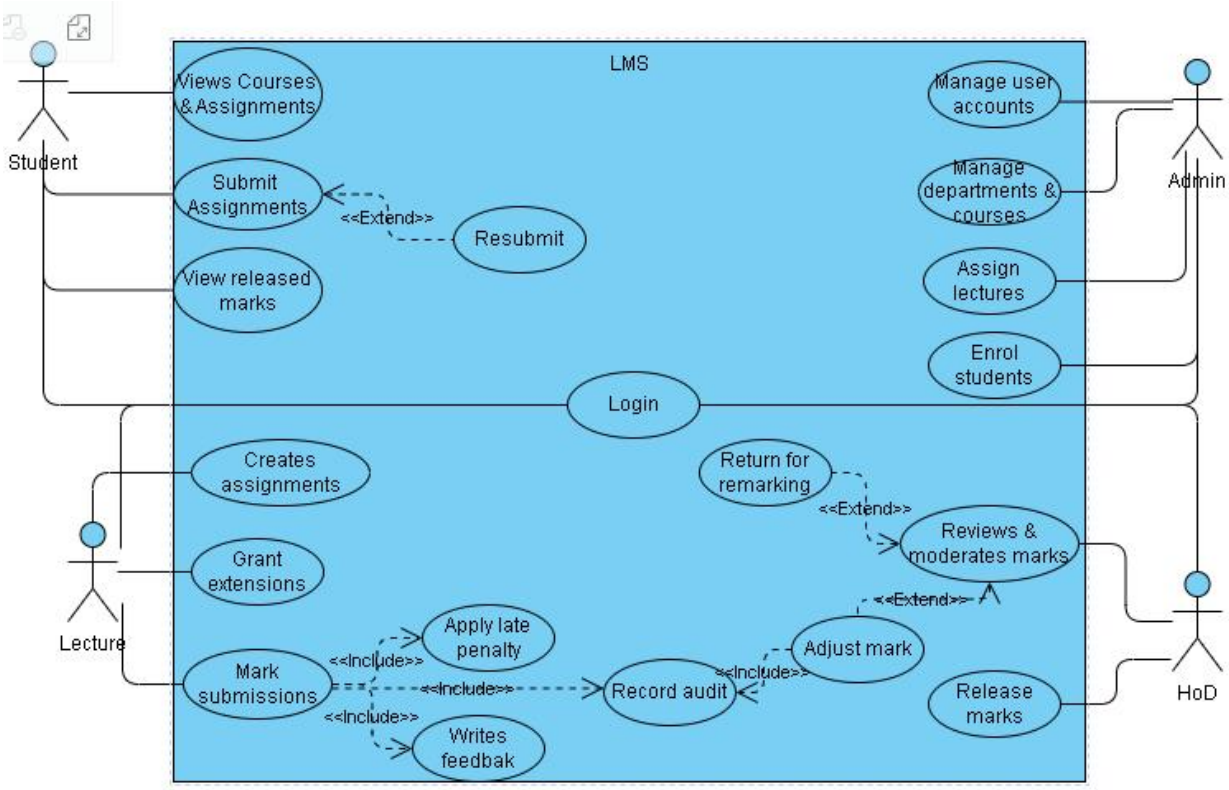
	belonging to those courses.		
FR-09	A student shall submit assignment work before or after the applicable deadline, subject to the system's submission rules.	Must	Test on-time, late and expired-deadline cases.
FR-10	A student shall resubmit assignment work before the applicable deadline.	Must	Test permitted and prohibited resubmissions.
FR-11	The system shall determine the applicable late penalty and calculate the resulting mark according to the approved rules.	Must	Test each late-penalty boundary.
FR-12	A lecturer shall assign a mark to a submission and provide written feedback.	Must	Test marking and feedback storage.
FR-13	The HoD shall view and moderate marks for courses within their department.	Must	Test departmental access and moderation.
FR-14	The system shall prevent students from viewing marks until the HoD has approved and released them.	Must	Test mark visibility before and after release.

Non-Functional Requirements

ID(NFR)	Requirement	Priority	Verification
NFR-01	The system shall complete a login attempt within 2 seconds when the database contains 800 registered student accounts and	Must	Measure login response time.

	representative lecturer, HoD and administrator accounts.		
NFR-02	The system shall store passwords only as salted hashes produced using a recognized password-hashing library; plain-text passwords shall not be stored.	Must	Inspect database and password-handling implementation.
NFR-03	The system shall operate fully offline (no internet connection required).	Must	System functions with network disabled.
NFR-04	The system shall prevent unauthorized access: a user of one role shall not be able to perform actions of another role.	Must	Role-based access tests.
NFR-05	The system shall remain responsive (UI response < 3 seconds) when managing data for 800 students.	Must	Performance test.
NFR-06	All mark changes shall be permanently recorded in an audit log that cannot be altered by normal users.	Must	Attempt to modify log fails.

2.3 Use Case Diagram



2.4 Fully Dressed Use Case Specifications

UC01 – Submit Assignment

Field	Description
Use Case ID	UC01
Use Case Name	Submit Assignment
Primary Actor	Student
Secondary Actors	None
Goal	Allow a student to submit (or resubmit) work for an assignment before the effective deadline.
Preconditions	1. Student is logged in. 2. Student is enrolled in the course that owns the assignment. 3. The assignment exists and is open for submission.

Field	Description
Postconditions	1. A new Submission record is created (or the existing one is updated if this is a resubmission). 2. Submission status is set to “Submitted” or “Late” according to the rules. 3. Submission timestamp is recorded.
Trigger	Student selects “Submit Assignment” from the menu.

Main Success Scenario (Basic Flow)

1. System displays the list of assignments for the student’s enrolled courses that are still open for submission.
2. Student selects an assignment.
3. System displays assignment details (title, description, deadline, maximum mark, any personal extension).
4. Student enters or uploads the submission content (text or file path, depending on implementation).
5. Student confirms submission.
6. System records the current date/time as the submission timestamp.
7. System determines whether the submission is on time or late (see Business Rules).
8. System saves the Submission with status “Submitted” or “Late”.
9. System confirms successful submission to the student.

Alternative Flows
A1 – Resubmission (before effective deadline)
At step 4, if a previous submission already exists:

- System informs the student that a previous version exists and asks for confirmation to overwrite.
- Student confirms.
- Continue from step 5. The previous content is replaced and the timestamp is updated.

A2 – Student has an individual extension
At step 7:

- System uses the extended deadline instead of the original deadline when calculating lateness.

Exception Flows
E1 – Assignment not found or not open
At step 2: System displays an error and returns to the assignment list.

E2 – Submission after the late window (more than 3 days late)
At step 7: System rejects the submission and informs the student that the late window has closed.
E3 – Student cancels
At any point before step 8: System discards the input and returns to the main menu.

Business Rules
BR1. A student may resubmit any number of times before the effective deadline (original deadline + any individual extension).
BR2. Late penalty is calculated later (at marking time) as 10% of the maximum mark for each full day late, up to a maximum of 3 days. After 3 days the mark will be zero.
BR3. The effective deadline = original deadline + individual extension days (if any).

UC02 – Mark Submission

Field	**Description**
Use Case ID	UC02
Use Case Name	Mark Submission
Primary Actor	Lecturer
Secondary Actors	None
Goal	Allow a lecturer to assign a numeric mark and written feedback to a student submission.
Preconditions	1. Lecturer is logged in. 2. Lecturer is assigned to the course that owns the assignment. 3. At least one submission exists for the assignment.
Postconditions	1. The submission has a numeric mark and written feedback. 2. Submission status is updated to “Marked”. 3. An audit log entry is created for the mark (even if this is the first mark).
Trigger	Lecturer selects “Mark Submissions” from the menu.

Main Success Scenario (Basic Flow)

1. System displays the list of assignments taught by the lecturer.
2. Lecturer selects an assignment.
3. System displays all submissions for that assignment (student name, submission date, current status).

4. Lecturer selects a submission.
5. System displays the submission content and any previous mark/feedback.
6. Lecturer enters a numeric mark (0 to maximum mark) and written feedback.
7. Lecturer confirms.
8. System applies any late penalty according to the business rules and stores the final mark.
9. System updates the submission status to “Marked”.
10. System writes an audit log entry (who, when, old mark, new mark, reason = “Initial marking” or “Remark”).
11. System confirms that the submission has been marked.

Alternative Flows
A1 – Remarking an already marked submission
At step 5: System shows the previous mark and feedback.
Lecturer enters new values → continue from step 7. Audit log records the change.

Exception Flows
E1 – Mark out of range
At step 6: System rejects the mark and asks the lecturer to re-enter a value between 0 and the maximum mark.
E2 – Lecturer cancels
At any point before step 8: No changes are saved.

Business Rules
BR1. Final mark = entered mark − late penalty (if applicable).
BR2. Late penalty = 10% of maximum mark × number of full days late (1–3). After 3 days late, final mark = 0.
BR3. Written feedback is mandatory.

UC03 – Moderate and Release Marks

Field	**Description**
Use Case ID	UC03
Use Case Name	Moderate and Release Marks
Primary Actor	Head of Department (HoD)
Secondary Actors	None
Goal	Allow the HoD to review, optionally change, and release marks for all courses in their department so that students can see them.

Field	Description
Preconditions	1. HoD is logged in. 2. HoD is assigned to a department. 3. There are submissions in status “Marked” for courses in that department.
Postconditions	1. Selected submissions have status “Released”. 2. Students can now view the mark and feedback. 3. Any mark changes made by the HoD are recorded in the audit log.
Trigger	HoD selects “Moderate Marks” from the menu.

Main Success Scenario (Basic Flow)

1. System displays the list of courses belonging to the HoD’s department.
2. HoD selects a course (or “All courses”).
3. System displays all submissions in status “Marked” for the selected course(s), showing student, assignment, lecturer mark, feedback, and calculated late penalty.
4. HoD reviews the list.
5. (Optional) HoD selects a submission and changes the mark and/or feedback.
6. HoD selects one or more submissions (or “Release All”).
7. HoD confirms release.
8. System updates the status of the selected submissions to “Released”.
9. If the HoD changed any mark, the system writes an audit log entry for each change.
10. System confirms that the marks have been released.

Alternative Flows
A1 – HoD changes a mark during moderation
At step 5: HoD enters a new mark → System records old value, new value, and reason (“Moderation”) in the audit log → continue.

Exception Flows
E1 – No marked submissions available
At step 3: System informs the HoD that there are no submissions ready for moderation.
E2 – HoD cancels
At any point before step 8: No status changes or mark changes are saved.

Business Rules
BR1. Only submissions in status “Marked” can be released.
BR2. Once released, the mark and feedback become visible to the student.
BR3. Every change of mark (including moderation changes) must be recorded in the immutable audit log.